Class 08: Breast Cancer Mini Project

Steven Studdard (PID: A19123470)

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Background

The goal of this mini-project is for us to explore a complete analysis using the unsupervised learning techniques overed in class.

Today we will analyze a biopsy data-set from a fine needle aspriation (FNA) of a breast mass.

Data Import

The data is made available as a CSV file for download. We can read this in using `read.csv()`:

```
fna.data <-  read.csv("WisconsinCancer.csv")

wisc.df <- data.frame(fna.data, row.names=1)

head(wisc.df, 4)
```

```
         diagnosis radius_mean texture_mean perimeter_mean area_mean
842302           M       17.99        10.38         122.80    1001.0
842517           M       20.57        17.77         132.90    1326.0
84300903         M       19.69        21.25         130.00    1203.0
```

```
84348301         M       11.42        20.38          77.58     386.1
         smoothness_mean compactness_mean concavity_mean concave.points_mean
842302           0.11840          0.27760         0.3001             0.14710
842517           0.08474          0.07864         0.0869             0.07017
84300903         0.10960          0.15990         0.1974             0.12790
84348301         0.14250          0.28390         0.2414             0.10520
         symmetry_mean fractal_dimension_mean radius_se texture_se perimeter_se
842302          0.2419                0.07871    1.0950     0.9053        8.589
842517          0.1812                0.05667    0.5435     0.7339        3.398
84300903        0.2069                0.05999    0.7456     0.7869        4.585
84348301        0.2597                0.09744    0.4956     1.1560        3.445
         area_se smoothness_se compactness_se concavity_se concave.points_se
842302    153.40      0.006399        0.04904      0.05373           0.01587
842517     74.08      0.005225        0.01308      0.01860           0.01340
84300903   94.03      0.006150        0.04006      0.03832           0.02058
84348301   27.23      0.009110        0.07458      0.05661           0.01867
         symmetry_se fractal_dimension_se radius_worst texture_worst
842302       0.03003             0.006193        25.38         17.33
842517       0.01389             0.003532        24.99         23.41
84300903     0.02250             0.004571        23.57         25.53
84348301     0.05963             0.009208        14.91         26.50
         perimeter_worst area_worst smoothness_worst compactness_worst
842302            184.60     2019.0           0.1622            0.6656
842517            158.80     1956.0           0.1238            0.1866
84300903          152.50     1709.0           0.1444            0.4245
84348301           98.87      567.7           0.2098            0.8663
         concavity_worst concave.points_worst symmetry_worst
842302            0.7119               0.2654         0.4601
842517            0.2416               0.1860         0.2750
84300903          0.4504               0.2430         0.3613
84348301          0.6869               0.2575         0.6638
         fractal_dimension_worst
842302                   0.11890
842517                   0.08902
84300903                 0.08758
84348301                 0.17300
```

```r
wisc.data <- wisc.df[,-1]

diagnosis <- as.factor(wisc.df$diagnosis)
```

Q1. How many observations are in this dataset?

```r
nrow(wisc.data)
```

```
[1] 569
```

> Q2. How many of the observations have a malignant diagnosis?

```r
table(wisc.df$diagnosis)["M"]
```

```
  M 
212 
```

> Q3. How many variables/features in the data are suffixed with _mean?

```r
length(grep("_mean",colnames(wisc.data)))
```

```
[1] 10
```

Principal Component Analysis (PCA)

We need to scale oru data before PCA with the `scale=TRUE` argument to `prcomp()`.

```r
wisc.pr <- prcomp(wisc.data, scale=T)
summary(wisc.pr)
```

```
Importance of components:
                          PC1    PC2     PC3     PC4     PC5     PC6     PC7
Standard deviation     3.6444 2.3857 1.67867 1.40735 1.28403 1.09880 0.82172
Proportion of Variance 0.4427 0.1897 0.09393 0.06602 0.05496 0.04025 0.02251
Cumulative Proportion  0.4427 0.6324 0.72636 0.79239 0.84734 0.88759 0.91010
                           PC8    PC9    PC10   PC11    PC12    PC13    PC14
Standard deviation     0.69037 0.6457 0.59219 0.5421 0.51104 0.49128 0.39624
Proportion of Variance 0.01589 0.0139 0.01169 0.0098 0.00871 0.00805 0.00523
Cumulative Proportion  0.92598 0.9399 0.95157 0.9614 0.97007 0.97812 0.98335
                          PC15    PC16    PC17    PC18    PC19    PC20   PC21
Standard deviation     0.30681 0.28260 0.24372 0.22939 0.22244 0.17652 0.1731
Proportion of Variance 0.00314 0.00266 0.00198 0.00175 0.00165 0.00104 0.0010
Cumulative Proportion  0.98649 0.98915 0.99113 0.99288 0.99453 0.99557 0.9966
                          PC22    PC23   PC24    PC25    PC26    PC27    PC28
Standard deviation     0.16565 0.15602 0.1344 0.12442 0.09043 0.08307 0.03987
```

```
Proportion of Variance 0.00091 0.00081 0.0006 0.00052 0.00027 0.00023 0.00005
Cumulative Proportion  0.99749 0.99830 0.9989 0.99942 0.99969 0.99992 0.99997
                          PC29    PC30
Standard deviation     0.02736 0.01153
Proportion of Variance 0.00002 0.00000
Cumulative Proportion  1.00000 1.00000
```

> Q4. From your results, what proportion of the original variance is captured by the first principal component (PC1)?

About 44% of the original variance is captured in PC1.

> Q5. How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

Up to PC3 is required to describe at least 70% of the original variance in the data.

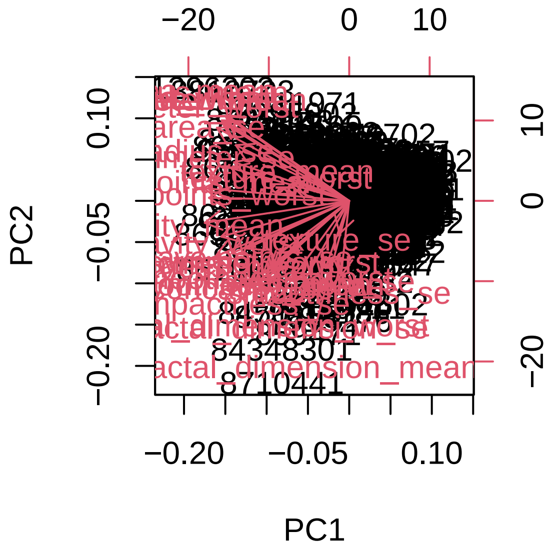
> Q6. How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

Up to PC7 is required to describe at least 90% of the original variance in the data.

> Q7. What stands out to you about this plot? Is it easy or difficult to understand? Why?

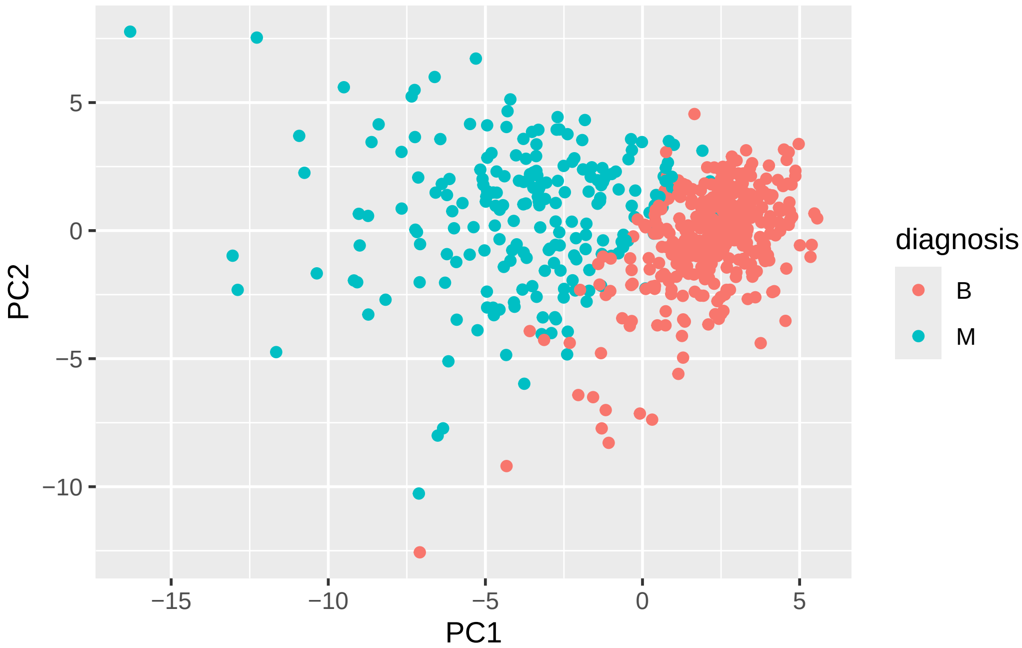
It stands out because its a massive cluster of information. The biplot is almost impossible to understand for me. It is just a large clutter of information without anyway to actually read it.

```r
biplot(wisc.pr)
```



Let’s see the “PC score plot”

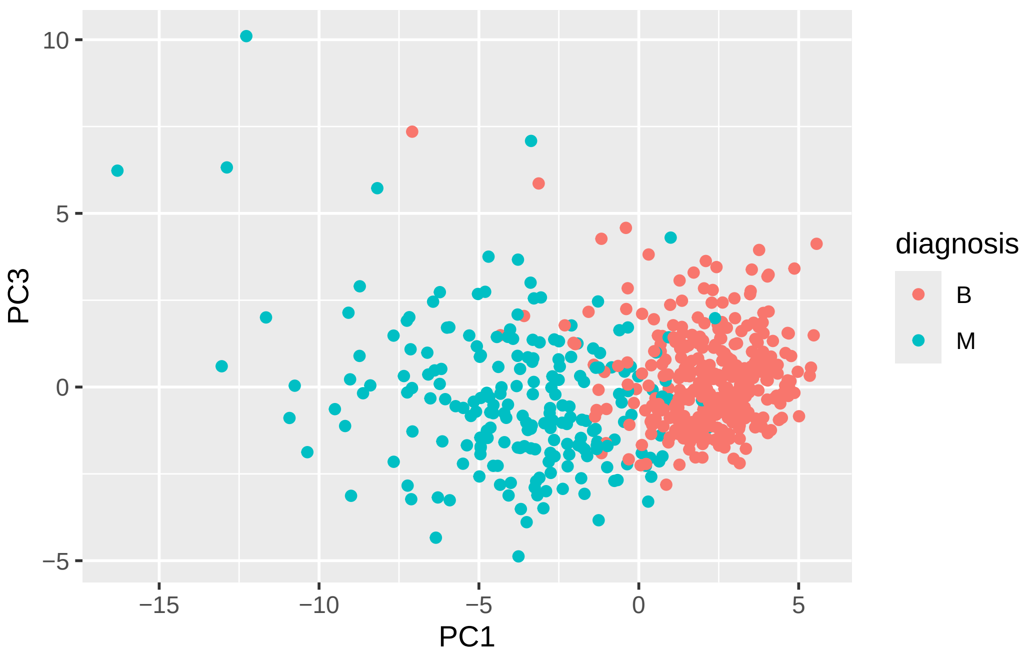
```
library(ggplot2)

ggplot(wisc.pr$x)+
  aes(PC1, PC2, col=diagnosis)+
  geom_point()
```



Q8. Generate a similar plot for principal components 1 and 3. What do you notice about these plots?

The two plots look quite similar other than the fact the scaling on the y axis where PC2/PC3 is different. There also seems to be more red dots overlapped with the blue than in the first PC2 plot.

```
ggplot(wisc.pr$x)+
  aes(PC1, PC3, col=diagnosis)+
  geom_point()
```



Checking Out Variance

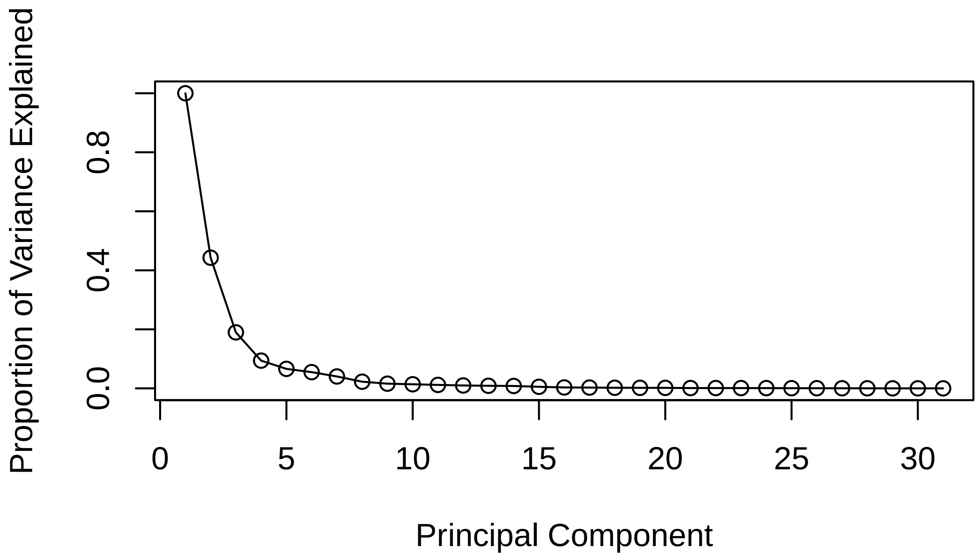
```r
attributes(wisc.pr)
```

```
$names
[1] "sdev"     "rotation" "center"   "scale"    "x"

$class
[1] "prcomp"
```

```r
pr.var <- wisc.pr$sdev^2
head(pr.var)
```

```
[1] 13.281608  5.691355  2.817949  1.980640  1.648731  1.207357
```

```r
# Variance explained by each principal component: pve
pve <- pr.var / sum(pr.var)

# Plot variance explained for each principal component
plot(c(1,pve), xlab = "Principal Component",
     ylab = "Proportion of Variance Explained",
     ylim = c(0, 1), type = "o")
```



Q9. For the first principal component, what is the component of the loading vector (i.e. wisc.pr$rotation[,1]) for the feature concave.points_mean? This tells us how much this original feature contributes to the first PC. Are there any features with larger contributions than this one?

```r
wisc.pr$rotation["concave.points_mean", 1]
```

```
[1] -0.2608538
```

```r
sort(abs(wisc.pr$rotation[,1]), decreasing = TRUE)
```

```
    concave.points_mean          concavity_mean    concave.points_worst
             0.26085376              0.25840048              0.25088597
       compactness_mean         perimeter_worst         concavity_worst
             0.23928535              0.23663968              0.22876753
           radius_worst          perimeter_mean              area_worst
             0.22799663              0.22753729              0.22487053
              area_mean             radius_mean            perimeter_se
             0.22099499              0.21890244              0.21132592
      compactness_worst               radius_se                 area_se
             0.21009588              0.20597878              0.20286964
      concave.points_se          compactness_se            concavity_se
```

```
         0.18341740              0.17039345              0.15358979
    smoothness_mean           symmetry_mean fractal_dimension_worst
         0.14258969              0.13816696              0.13178394
   smoothness_worst          symmetry_worst           texture_worst
         0.12795256              0.12290456              0.10446933
       texture_mean    fractal_dimension_se  fractal_dimension_mean
         0.10372458              0.10256832              0.06436335
        symmetry_se              texture_se           smoothness_se
         0.04249842              0.01742803              0.01453145
```

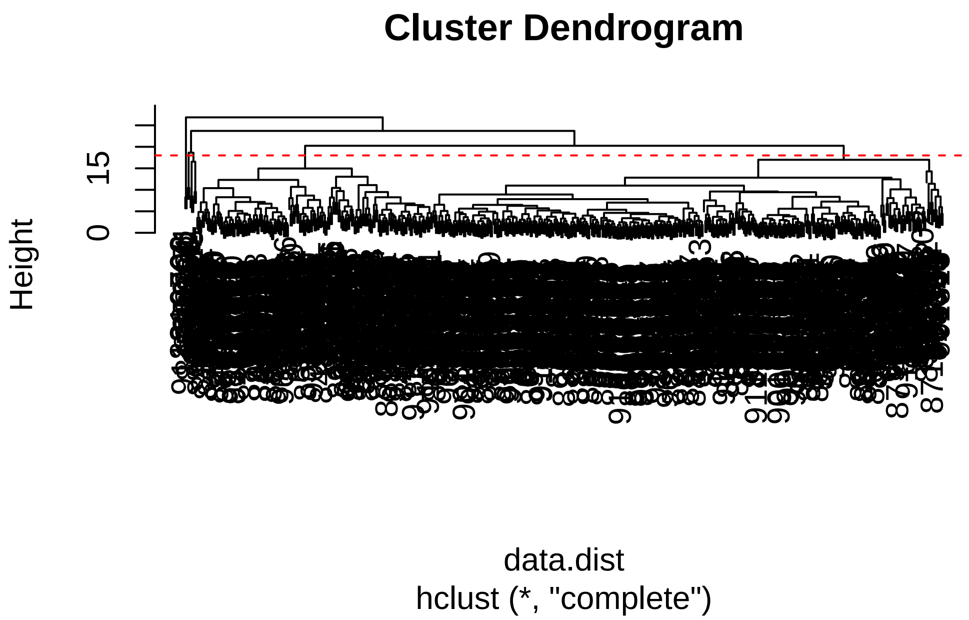
H Clust

```
data.scaled <- scale(wisc.data)
data.dist <- dist(data.scaled)
wisc.hclust <- hclust(data.dist, method="complete")
```

> Q10. Using the plot() and abline() functions, what is the height at which the clustering model has 4 clusters?

Looking at the plot below it seems that around height 18 it would create 4 clusters.

```
plot(wisc.hclust)
abline(h=18, col="red", lty=2)
```




```r
wisc.hclust.clusters <- cutree(wisc.hclust, k=4)
table(wisc.hclust.clusters, diagnosis)
```

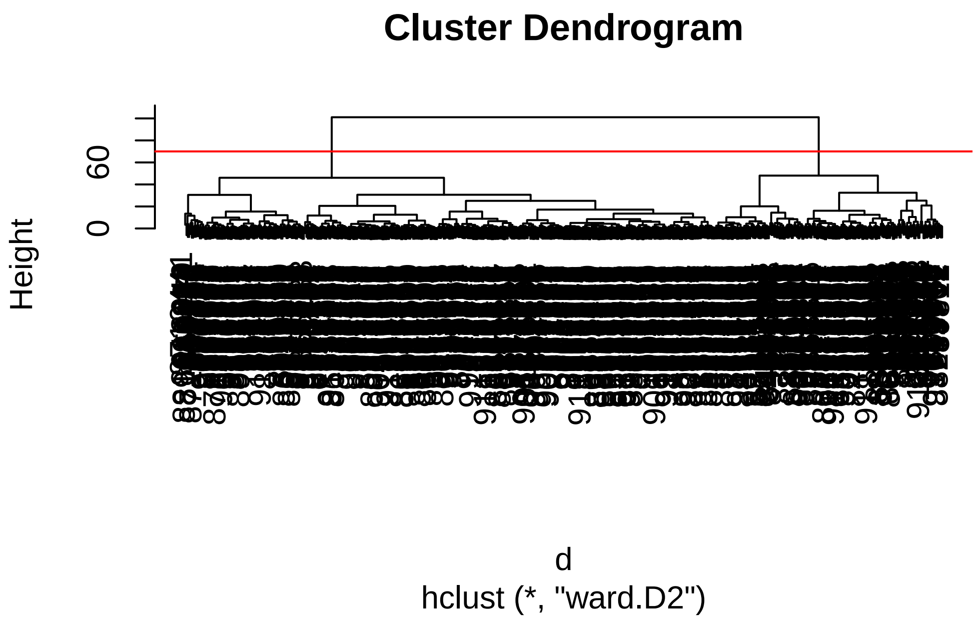
```
                    diagnosis
wisc.hclust.clusters   B   M
                   1  12 165
                   2   2   5
                   3 343  40
                   4   0   2
```

> Q12. Which method gives your favorite results for the same data.dist dataset? Explain your reasoning.

I think I prefer looking at the hclust plot tree from the wisc.hclust data set we made from data.dist. It gives a much more visible tree to create an abline and decine the amount of clusters. It gives the same information, but in a cleaner format.

Combining Methods

```
d<-dist(wisc.pr$x[,1:4])
wisc.pr.hclust <- hclust(d, method="ward.D2")
plot(wisc.pr.hclust)
abline(h=70, col="red")
```




```
grps <- cutree(wisc.pr.hclust, h=70)
table(grps)
```

```
grps
  1   2
171 398
```

How does tihs clustering `grps` correspond to the expert `diagnosis`?

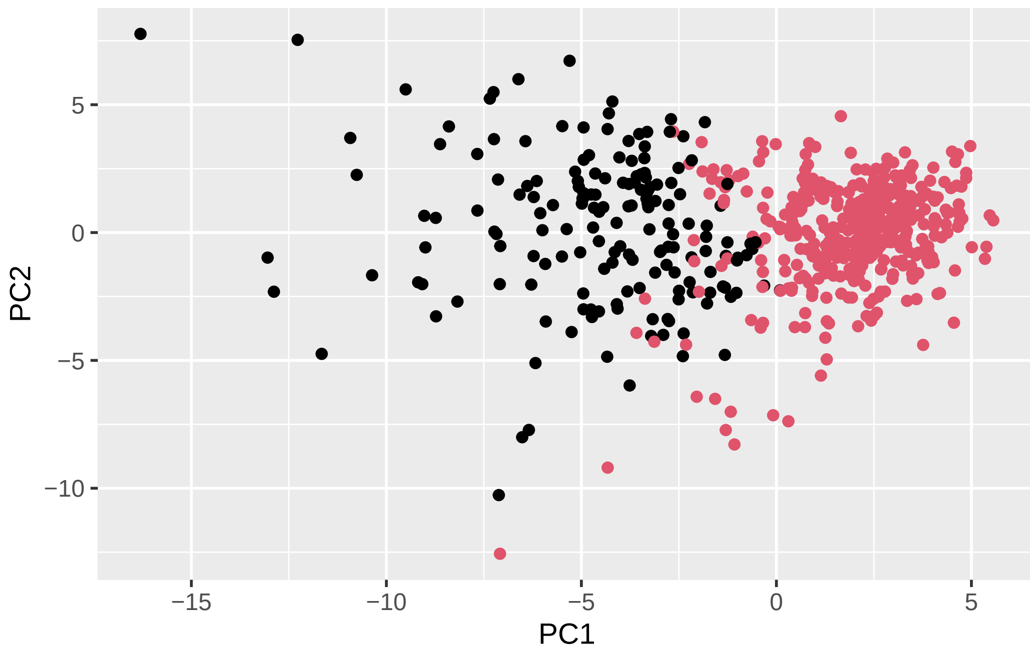
```
table(diagnosis)
```

```
diagnosis
  B   M
357 212
```

```r
table(diagnosis, grps)
```

```
         grps
diagnosis   1   2
        B   6 351
        M 165  47
```

```r
ggplot(wisc.pr$x) +
  aes(PC1, PC2) +
  geom_point(col=grps)
```



> Q13. How well does the newly created hclust model with two clusters separate out the two "M" and "B" diagnoses?

I'd say that it does do better. I say this because instead of having a tree with 4 clusters, we can instead see on this plot above that there is two distinct clusters.

> Q14. How well do the hierarchical clustering models you created in the previous sections (i.e. without first doing PCA) do in terms of separating the diagnoses? Again, use the table() function to compare the output of each model (wisc.hclust.clusters and wisc.pr.hclust.clusters) with the vector containing the actual diagnoses.

Looking at the table below. I can generally say the mixed method is closer to the expert diagnosis than the hclust method alone. Both methods are relatively close, but pr.hclust is just slightly more alligned with the experts.

```
table(wisc.hclust.clusters, diagnosis)
```

```
                    diagnosis
wisc.hclust.clusters   B   M
                   1  12 165
                   2   2   5
                   3 343  40
                   4   0   2
```

```
##Grps is same as wisc.pr.hclust.clusters
table(grps, diagnosis)
```

```
    diagnosis
grps   B   M
   1   6 165
   2 351  47
```

```
table(diagnosis)
```

```
diagnosis
  B   M
357 212
```

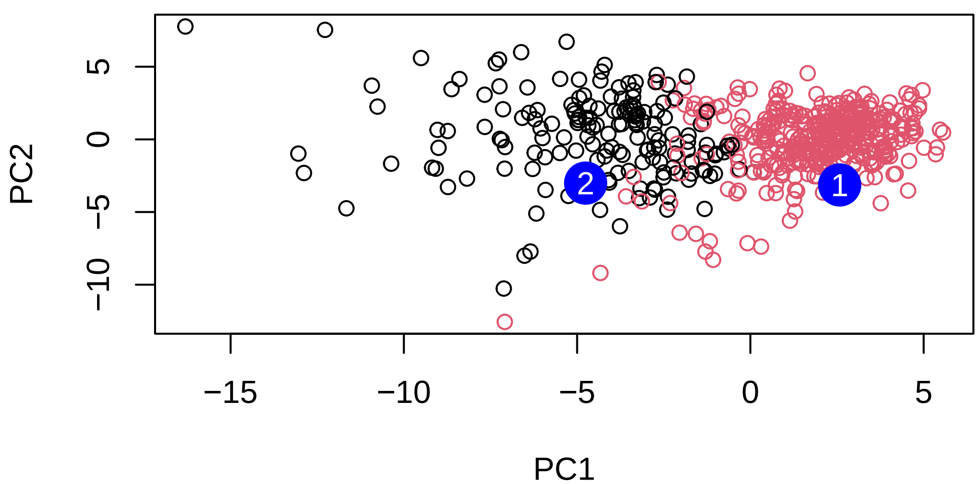
Prediction

```
#url <- "new_samples.csv"
url <- "https://tinyurl.com/new-samples-CSV"
new <- read.csv(url)
npc <- predict(wisc.pr, newdata=new)
npc
```

```
           PC1       PC2        PC3        PC4       PC5        PC6        PC7
[1,]  2.576616 -3.135913  1.3990492 -0.7631950  2.781648 -0.8150185 -0.3959098
[2,] -4.754928 -3.009033 -0.1660946 -0.6052952 -1.140698 -1.2189945  0.8193031
```

```
             PC8        PC9       PC10       PC11       PC12      PC13     PC14
[1,] -0.2307350 0.1029569 -0.9272861 0.3411457  0.375921 0.1610764 1.187882
[2,] -0.3307423 0.5281896 -0.4855301 0.7173233 -1.185917 0.5893856 0.303029
          PC15       PC16        PC17        PC18        PC19       PC20
[1,] 0.3216974 -0.1743616 -0.07875393 -0.11207028 -0.08802955 -0.2495216
[2,] 0.1299153  0.1448061 -0.40509706  0.06565549  0.25591230 -0.4289500
           PC21       PC22       PC23       PC24        PC25         PC26
[1,]  0.1228233 0.09358453 0.08347651  0.1223396  0.02124121  0.078884581
[2,] -0.1224776 0.01732146 0.06316631 -0.2338618 -0.20755948 -0.009833238
             PC27        PC28         PC29         PC30
[1,]  0.220199544 -0.02946023 -0.015620933  0.005269029
[2,] -0.001134152  0.09638361  0.002795349 -0.019015820
```

```r
plot(wisc.pr$x[,1:2], col=grps)
points(npc[,1], npc[,2], col="blue", pch=16, cex=3)
text(npc[,1], npc[,2], c(1,2), col="white")
```



Q16. Which of these new patients should we prioritize for follow up based on your results?

We should follow up with patient 2 as it seems that they are more likely to have a malignant tumor when i view the predicted results.